

# Communication Lower Bounds for Multi-Party Graph Traversal

## Quarter 2 Project Report

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## Abstract

TBD

Repository: <https://github.com/rybplayer/DSCCapstone>

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# 1 Introduction

This report studies communication lower and upper bounds for multi-party graph traversal problems, motivated by real systems in which no single agent holds the full network state. We focus on settings where a global graph is known but each party privately owns a subset of vertices and edges, and the task is to determine reachability or shortest paths that must lie within the union of their private subgraphs.

## 1.1 Motivation from multi-party traversal case studies

Large-scale transportation and infrastructure disruptions create a natural model for multi-party graph traversal. Consider the 2010 European volcanic ash shutdown: air-navigation service providers, airports, and airlines each control disjoint portions of airspace and schedules. Rerouting passengers is a shortest-path problem in a layered graph whose feasible vertices and edges change over time. No party can broadcast its full real-time state, both for operational and privacy reasons, so route feasibility must be computed with limited information exchange.

Similar graph-traversal constraints appear in urban transit shutdowns and phased restorations, where distinct agencies own stations and tunnels. After Hurricane Sandy, service availability evolved over weeks, and agencies coordinated alternate routes under partial knowledge of what infrastructure was open. Highway detours after the I-95 bridge collapse in Philadelphia illustrate the same phenomenon in a road network: different operators own highway segments and toll roads, and a single failed cut edge forces traffic onto alternative paths with asymmetric knowledge of capacities.

The maritime and logistics examples are equally instructive. The 2021 Suez Canal blockage forced carriers to reroute around the Cape of Good Hope or wait, a decision that depends on schedule and capacity information held privately by shipping lines and port operators. When the Baltimore Key Bridge collapsed, access to the port was blocked and cargo was diverted to alternate ports, again requiring coordination across a sea–port–land graph with private operational constraints.

Communication bottlenecks become even sharper in critical infrastructure networks. During the 2021 Texas ERCOT blackout, grid operators and utilities effectively solve a constrained flow and connectivity problem as generation and transmission nodes go offline. The Colonial Pipeline cyber disruption similarly forced fuel logistics to reroute through alternate depots and terminals with incomplete visibility. Finally, the 2021 Meta backbone withdrawal and the 2016 Dyn DNS attack show that digital services rely on reachability in routing and dependency graphs, where policy and security constraints limit information sharing across administrative domains.

These cases motivate a theoretical study of multi-party graph traversal: agents must collaboratively decide reachability and shortest paths while their private subgraphs, capacities, or weights are not fully shareable. The fundamental question is how much communication is necessary, and how randomized protocols, sketches, or partial disclosures can reduce the

burden while preserving correctness.

## 1.2 Why communication is the bottleneck

In the model we consider, each party has unlimited local computation but limited ability to share data. This is realistic: air traffic control centers, network operators, or utilities can compute detailed local routes, but cannot afford to broadcast all local states and constraints due to bandwidth, privacy, regulatory, or security limitations. The communication cost, not the local computation, is what constrains real-time coordination.

Communication complexity makes this separation explicit. In particular, it shows that even when each party can compute arbitrarily powerful local summaries, some global questions still require a large number of bits to be exchanged. For example, deciding whether two privately held sets intersect already requires linear communication in the worst case. This lower bound appears again in many traversal problems via reductions that embed set disjointness or equality into graph reachability. Thus, communication is not simply a practical nuisance: it is an inherent barrier that must be designed around.

## 1.3 Why bounds matter for hardware, networks, and algorithms

Lower and upper bounds provide practical guidance in three ways. First, they inform hardware design. If a reachability or shortest-path query provably requires  $\Omega(n)$  bits of communication, then any system intended to answer such queries at scale must provision sufficient bandwidth or memory to move that information. Second, bounds guide network architecture: if communication is unavoidable, the topology of communication links, the placement of caches, and the choice of aggregation points become part of the algorithmic design. Third, bounds shape algorithm development: a tight lower bound rules out overly ambitious protocols and motivates either approximate solutions, randomized protocols with small error, or problem relaxations that admit sublinear communication. Conversely, upper bounds show which tasks are feasible under realistic constraints, making them valuable to practitioners who must trade correctness, latency, and privacy.

## 1.4 Communication complexity basics

We briefly formalize the model. Let  $f : \mathcal{X} \times \mathcal{Y} \rightarrow \{0, 1\}$  be a function. In the two-party model, Alice receives  $x \in \mathcal{X}$  and Bob receives  $y \in \mathcal{Y}$ . A deterministic protocol is a binary decision tree whose nodes are labeled by which party speaks and whose edges are labeled by bits; the transcript is the sequence of bits exchanged. The protocol must output  $f(x, y)$  at a leaf. The deterministic communication complexity  $D(f)$  is the minimum, over all correct protocols, of the maximum transcript length over all inputs.

Randomized protocols allow parties to use random coins to reduce communication. A public-coin protocol assumes a shared random string, while a private-coin protocol lets

each party sample independently. Fixing the randomness yields a deterministic protocol, so a randomized protocol is a distribution over deterministic protocols. The randomized communication complexity  $R_\epsilon(f)$  is the minimum number of communicated bits needed to compute  $f$  with error at most  $\epsilon$  on every input. Public-coin and private-coin models are equivalent up to small overhead, and randomness often trades a small probability of error for large savings in communication.

Two classical benchmark problems are equality and disjointness. Equality asks whether  $x = y$  for  $x, y \in \{0, 1\}^n$ , and disjointness asks whether  $X \cap Y = \emptyset$  for subsets of  $[n]$ . Both are complete for a wide range of lower-bound techniques. Deterministically, equality requires  $n$  bits in the worst case, and disjointness requires  $\Omega(n)$  communication. Randomization yields large gains for equality via hashing, but not for disjointness, which remains hard.

## 1.5 From communication to graph traversal

Graph traversal problems can be embedded with these benchmarks. A simple variant illustrates the connection. Let a known undirected graph  $G = (V, E)$  be given. Alice receives a start vertex  $v$  and a subset  $X \subseteq V$ , Bob receives an end vertex  $w$  and a subset  $Y \subseteq V$ . The task is to output 1 if there exists a path from  $v$  to  $w$  using only vertices in  $X \cup Y$  and edges that are wholly contained within  $X$  or wholly contained within  $Y$ . By constructing a graph in which each bit of an input string controls which local vertices are present, one can reduce equality to this traversal problem. This implies that even this restricted reachability task has deterministic communication complexity  $\Omega(n)$  when  $|V| = \Theta(n)$ .

The case studies above map naturally to this model. Each actor controls a subgraph of the global infrastructure; edges represent feasible transitions, and reachability encodes whether a route exists that respects ownership and availability constraints. The reduction perspective explains why coordination can be intrinsically expensive: some instances encode hard communication problems, so no protocol can be both exact and frugal with communication in the worst case.

## 1.6 Scope of this report

The remainder of this report develops a communication-complexity view of multi-party graph traversal. We formalize traversal models, establish lower bounds via reductions, and explore upper bounds that exploit randomness, sketches, or structure. Throughout, the guiding message is that communication is not merely an implementation detail. It is a fundamental resource with provable limits, and understanding those limits yields actionable insights for designing distributed routing, resilient infrastructure, and scalable coordination protocols.

## References